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✅ Data Filtering

```python
import pandas as pd
```

```python
data1 = {
    "Name" : ["ABC","DEF","GHI","JKL"],
    "Marks" : [90,67,85,90]
}
df1 = pd.DataFrame(data1)
```

```python
print(df1)
```

```
  Name  Marks
0  ABC     90
1  DEF     67
2  GHI     85
3  JKL     90
```

```python
print(df1["Marks"]>70)
```

```
0     True
1    False
2     True
3     True
Name: Marks, dtype: bool
```

```python
print(df1[df1["Marks"]>70])
```

```
  Name  Marks
0  ABC     90
2  GHI     85
3  JKL     90
```

Data Sorting

```python
# sort values function = sort_values()
```

```python
df2 = pd.DataFrame(data1)
df2.sort_values("Marks") # ascending = True
```

	Name	Marks
1	DEF	67
2	GHI	85
0	ABC	90
3	JKL	90

```python
df2 = pd.DataFrame(data1)
df2.sort_values("Marks", ascending = False) # descending -> ascending = False
```

	Name	Marks
0	ABC	90
3	JKL	90
2	GHI	85
1	DEF	67

✅ Checking Missing Value

```python
import numpy as np
data3 ={
    "Name" : ["ABC","DEF","GHI","JKL"],
    "Marks" : [90,67,np.nan,90]
}
df3 = pd.DataFrame(data3)
print(df3)
```

```
  Name  Marks
0  ABC   90.0
1  DEF   67.0
2  GHI    NaN
3  JKL   90.0
```

```python
# Identifying missing values
df3.isnull()
```

	Name	Marks
0	False	False
1	False	False
2	False	True
3	False	False

Removing Missing Value

```
data4 ={
    "Name" : ["ABC","DEF","GHI","JKL"],
    "Marks" : [90,67,np.nan,90]
}
df4 = pd.DataFrame(data4)
print(df4)
```

```
  Name  Marks
0  ABC   90.0
1  DEF   67.0
2  GHI    NaN
3  JKL   90.0
```

```
# Removing missng value
df4.dropna()
```

	Name	Marks
0	ABC	90.0
1	DEF	67.0
3	JKL	90.0

✅ Filling Missing Value

```python
# Filling missing values
df3.fillna(0, inplace = True)
```

	Name	Marks
0	ABC	90.0
1	DEF	67.0
2	GHI	0.0
3	JKL	90.0

```python
# Replacing by mean
data5 ={
    "Name" : ["ABC","DEF","GHI","JKL"],
    "Marks" : [90,67,np.nan,90]
}
df5 = pd.DataFrame(data5)
df5.fillna(df5["Marks"].mean())
```

	Name	Marks
0	ABC	90.000000
1	DEF	67.000000
2	GHI	82.333333
3	JKL	90.000000

```python
# Replacing by unknown
data6 ={
    "Name" : ["ABC","DEF","GHI","JKL"],
    "Marks" : [90,67,np.nan,90]
}
df6 = pd.DataFrame(data6)
df6.fillna("Unknown")
```

	Name	Marks
0	ABC	90.0
1	DEF	67.0
2	GHI	Unknown
3	JKL	90.0

✅ Group-By Operation

```
# Group by operation
```

```
emp = {
    "dept" : ["HR","IF","HR","IF"],
    "salary" : [10000,20000,15000,23000]
}
empDF = pd.DataFrame(emp)
```

```
print(empDF)
```

```
  dept  salary
0   HR   10000
1   IF   20000
2   HR   15000
3   IF   23000
```

```
empDF.groupby("dept")["salary"].mean()
```

```
dept
HR    12500.0
IF    21500.0
Name: salary, dtype: float64
```